

FEEDFORWARD NETWORKS, LOSS FUNCTIONS & REGULARIZATION

AI506 — Advanced Machine Learning

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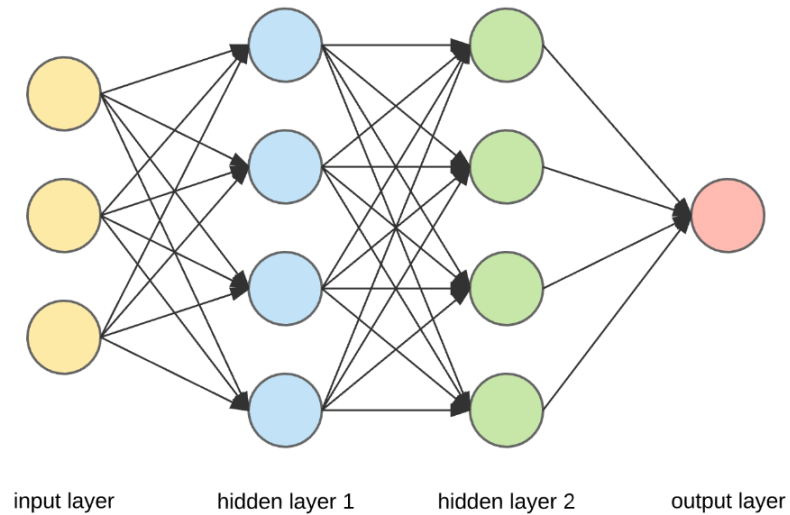


Figure 1: Feedforward net illustration

$$f(x) = \sigma(Wx + b)$$

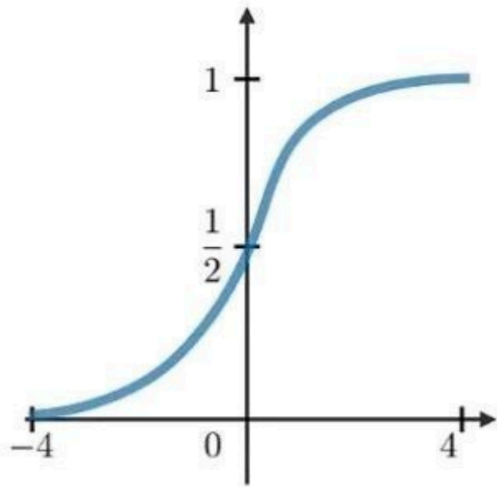
Let $f : X \rightarrow Y$

Definition: Universal Approximation Theorem

For any compact continuous function f , an approximation function $\hat{f}$ and an $\varepsilon > 0$, then

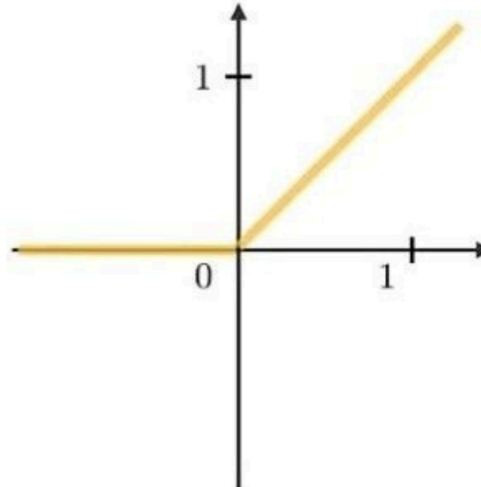
$$|f(x) - \hat{f}(x)| < \varepsilon \quad \forall x \in \mathcal{X}$$

Sigmoid



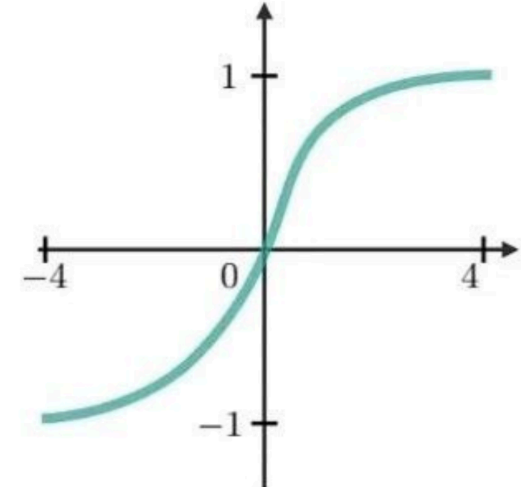
$$\sigma(z) = \frac{1}{1+e^{-z}}$$

ReLU



$$\sigma(z) = \max(0, z)$$

Tanh



$$\sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Mean Squared Error

$$\ell = (\hat{y} - y)^2$$

$$\mathcal{L} = \frac{1}{N} \sum_i (\hat{y}_i - y_i)^2$$

Cross Entropy

$$\ell = -y \log(h_{\theta}(x)) \quad \text{where } y \in \{0, 1\}$$

$$\ell_c = -\sum_c y_c \log(h_{\theta}(x_c))$$

$$\mathcal{L} = -\sum_i \sum_c y_{i,c} \log(h_{\theta}(x_{i,c}))$$

Binary Cross Entropy

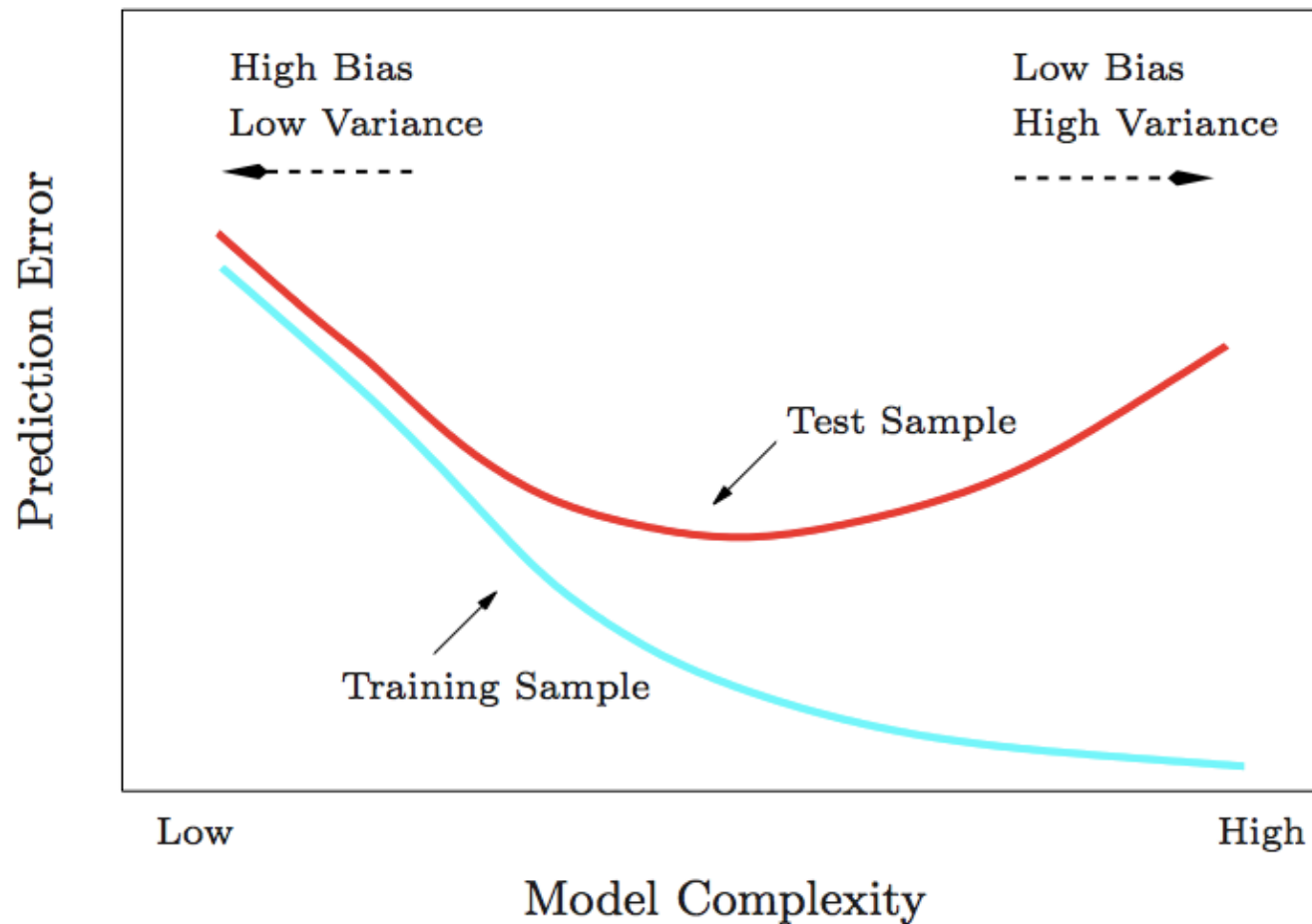
$$\mathcal{L} = - \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

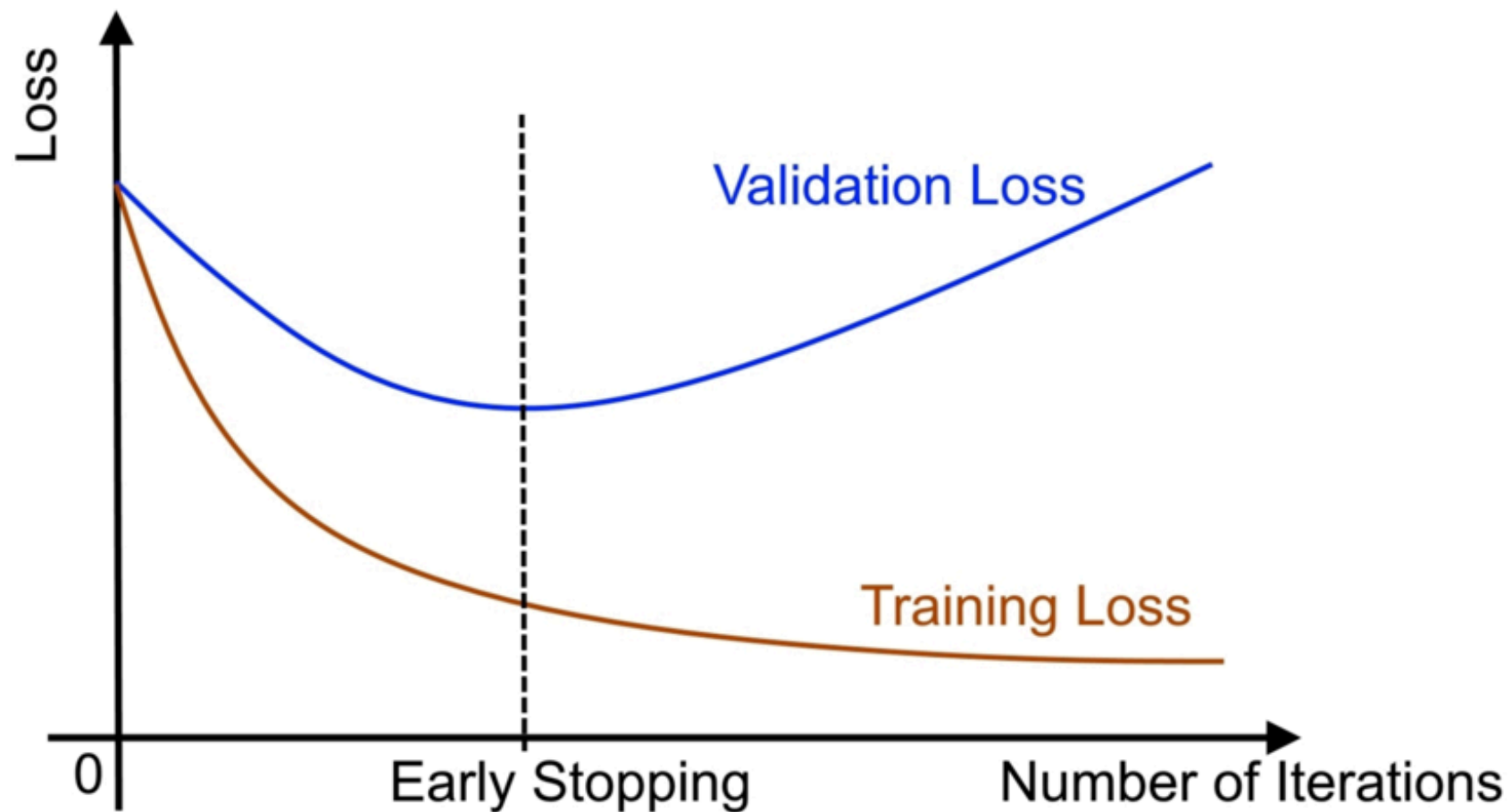
Maximum Likelihood Estimation

$$\hat{\theta} = \arg \max_{\theta} \underbrace{\sum_i \log p(\hat{y}_i \mid x_i, \theta)}_{-\mathcal{L}}$$

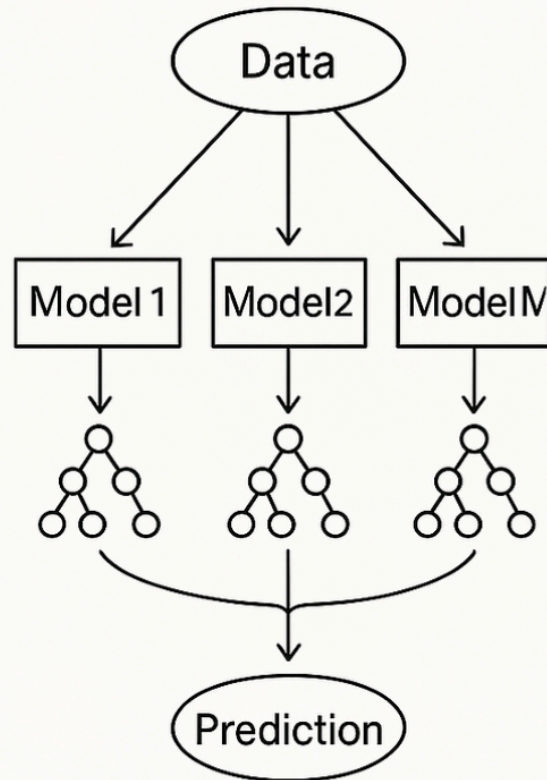
$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\theta) = \arg \min_{\theta} - \sum_i \log p(\hat{y} \mid x_i, \theta)$$

$$\begin{aligned}\mathcal{L}_{\text{MLE}} &= - \sum_i \log p(y_i \mid x_i, \theta) \\ &= - \sum_i \sum_c \log p(y_{i,c} \mid x_i, \theta) \\ &= - \sum_i \log \prod_c h_\theta(x_{i,c}) \quad \text{since } y_{i,c} \in \{0, 1\} \\ &= - \sum_i \sum_c y_{i,c} \log h_\theta(x_{i,c}) \\ &= \mathcal{L}_{\text{CE}}\end{aligned}$$

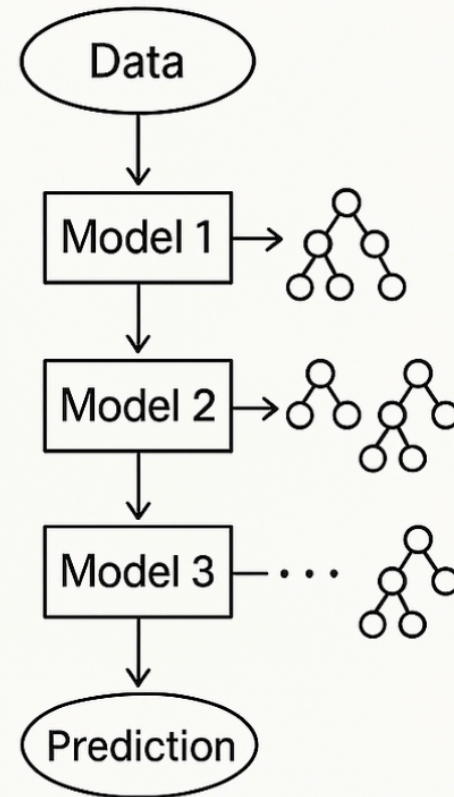




Bagging



Boosting



L1 (Absolute)

$$\|w\|_{L1} = \lambda \sum_i |w_i|$$

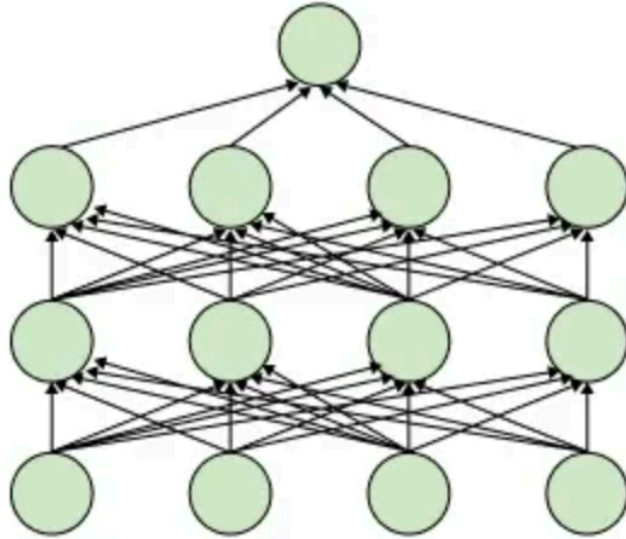
L2 (Squared)

$$\|w\|_{L2} = \lambda \sum_i w_i^2$$

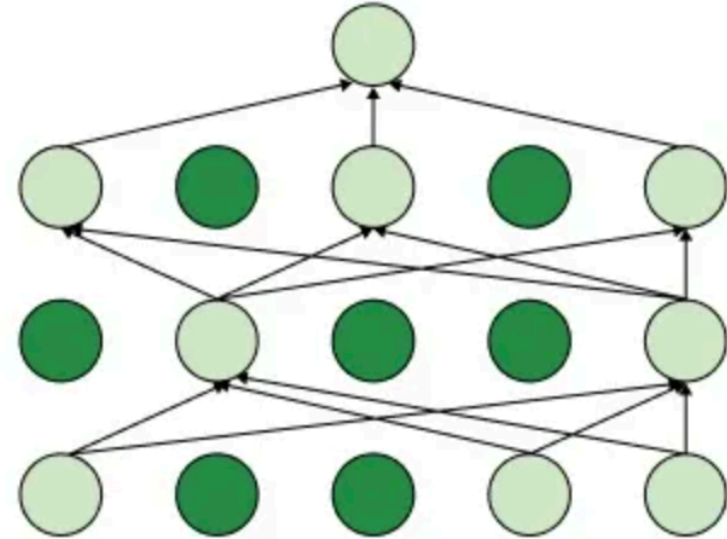
Let λ be a weight decay parameter and α be the stepsize for GD. Lets apply L2.

$$w' = (1 - \lambda\alpha)w - \alpha \nabla \mathcal{L}(w, X, y)$$

A neural network before dropout

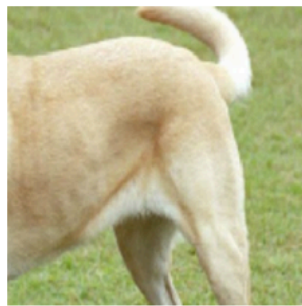


A neural network after dropout





(a) Original



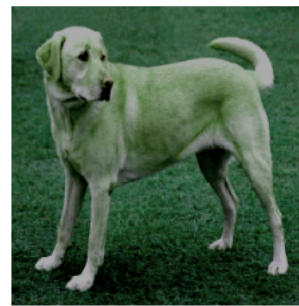
(b) Crop and resize



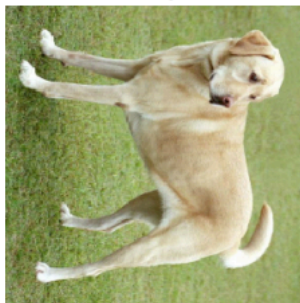
(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering